Appendix 20

Proof of the disentanglement lemma

In this appendix we present a proof of the crucial identity (9.80), i.e.

$$e^{\hat{A}+\hat{B}} = e^{-\frac{1}{2}[\hat{A},\hat{B}]}e^{\hat{A}}e^{\hat{B}}\,, \tag{A20.1}$$

under the assumption that $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$ and $\hat{B}$.

Let us define

$$\hat{B}(\lambda) \equiv e^{-\lambda\hat{A}}\hat{B}e^{\lambda\hat{A}}\,, \tag{A20.2}$$

where λ is a real number in the range $0 \le \lambda \le 1$. $\hat{B}(\lambda)$ satisfies the "equation of motion"

$$\frac{d\hat{B}(\lambda)}{d\lambda} = -e^{-\lambda\hat{A}}[\hat{A}, \hat{B}]e^{\lambda\hat{A}} = -[\hat{A}, \hat{B}]\,, \tag{A20.3}$$

where we have used the fact that $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$. Notice that the right–hand side of this equation is independent of λ: thus, integrating over λ, and noting that $\hat{B}(0) \equiv \hat{B}$, we arrive at the important result

$$\hat{B}(\lambda) = \hat{B} - \lambda[\hat{A}, \hat{B}]\,. \tag{A20.4}$$

As a consequence the two operators $\hat{B}(\lambda)$ and $\hat{B}(\lambda')$ commute with each other for arbitrary values of λ and λ'. Next consider the operator

$$\hat{U}(\lambda) \equiv e^{-\lambda\hat{A}}e^{\lambda(\hat{A}+\hat{B})}\,. \tag{A20.5}$$

It is easy to verify that $\hat{U}(\lambda)$ satisfies the equation of motion

$$\frac{d\hat{U}(\lambda)}{d\lambda} = \hat{B}(\lambda)\hat{U}(\lambda)\,. \tag{A20.6}$$

The formal solution of this equation is

$$\hat{U}(\lambda) = T_\lambda e^{\int_0^\lambda \hat{B}(\lambda')d\lambda'}\,, \tag{A20.7}$$

where T_λ arranges the operators $\hat{B}(\lambda)$ in order of decreasing values of λ (see Section 6.2.2, where a similar equation is solved to obtain the time-evolution operator). In this case, the T_λ operator is however unnecessary, for, as we have just seen, the operators $\hat{B}(\lambda)$ with different values of λ always commute with each other. Thus, the solution for $\hat{U}(\lambda)$,

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evaluated at $\lambda = 1$, is simply given by

$$\hat{U}(1) = e^{\int_0^1 (\hat{B} - \lambda' [\hat{A}, \hat{B}]) d\lambda'} \\ = e^{\hat{B}} \, e^{-\frac{1}{2}[\hat{A}, \hat{B}]} \, , \tag{A20.8}$$

where we have used Eq. (A20.4) and the fact that $[\hat{A}, \hat{B}]$ commutes with $\hat{B}$. Notice that, by the definition of $\hat{U}(\lambda)$ in Eq. (A20.5), we also have

$$\hat{U}(1) = e^{-\hat{A}} \, e^{\hat{A}+\hat{B}} \, . \tag{A20.9}$$

Thus, comparing Eqs. (A20.8) and (A20.9), we arrive at Eq. (A20.1).

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